Total Error: What Classifier Validation Establishes About Downstream Estimates

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Abstract

Researchers often estimate quantities from predicted labels: a prevalence, a difference between groups, or a regression coefficient. When a label is reused across records, its mistake can affect an estimate many times; a confusion matrix records how many mistakes occurred but loses their locations. This note asks what validation can establish about a fixed estimate built from such labels. For estimates linear in the labels, I give sharp bounds using exact confusion counts and show when those counts identify the estimate. A probability audit of target items yields a simultaneous count-based confidence certificate, while a direct audit retains the weighted residuals needed for a particular estimate. In a browsing-data stress test, exact error counts leave the signs of 84 group comparisons unresolved. A separate comparison of adult-domain classifiers shows similar information loss with observed disagreements, without treating either classifier as truth. I also distinguish these fixed-label results from score plug-ins, which need relevant conditional mean assumptions, and from regression with aggregated predicted covariates, whose coefficient is nonlinear in the labels. The practical choice is what to validate, and how much of that evidence to retain, for the estimate the researcher intends to report.

Suppose a researcher classifies websites, documents, or other items, then sums those labels across people or records. The reported quantity might be a total, a difference in means, or a coefficient from regressing an aggregate outcome on observed covariates. The same label may enter many records, and its influence depends on the particular quantity being estimated. Accuracy and the usual false-positive and false-negative rates count items; they do not record which influential items were misclassified. Knowing the exact confusion matrix for the target collection therefore need not determine even the sign of a group comparison.

The distinction between classifying items well and estimating an aggregate well is established. Work on quantification addresses class prevalence (Hopkins and King, 2010; Forman, 2008; Vaz et al., 2019); cost quantification already allows nonnegative item-specific weights (Forman, 2008). Design-based supervised learning corrects aggregates using probability samples of verified labels (Egami et al., 2023). Election and financial auditing use random checks to control weighted error (Stark, 2008; Shekhar et al., 2023). Machine-generated covariates, including indices assembled from classifications, are studied in regression (Fong and Tyler, 2021; Battaglia et al., 2025; Kluger et al., 2025). These results motivate a narrower question: for one declared estimate and a fixed set of target items, what does a particular validation summary actually establish? Reused labels and signed comparison weights make that question especially consequential.

*gsood07@gmail.com. The argument that classification errors in browsing totals depend on domain popularity appeared in my September 2018 note, linked by an October 2018 repository commit. It also appeared in the June 2020 manuscript coauthored with Suriyan Laohaprapanon (Laohaprapanon and Sood, 2020). The standalone note and original simulation were added in June 2023. This revision corrects the error-rate definitions and develops the validation bounds and audit results. Code and history: `https://github.com/finite-sample/total_error`.

I answer it in three steps. First, I map each linear estimate back to its item weights and derive the smallest interval compatible with exact target-item confusion counts. I characterize when those counts are sufficient. Second, I show how a probability audit gives a simultaneous count certificate and why a direct weighted residual audit can retain more information for a declared contrast. Third, I apply the calculations to a browsing-data stress test and to observed disagreements among adult-domain classifiers. The latter is an information exercise because neither classification is adjudicated truth. Aggregated predicted covariates and calibrated scores are related uses of the same labels, but interpreting a regression slope or a score plug-in requires additional assumptions. The note offers a way to match validation evidence to the downstream estimand; it does not introduce a new classifier or a new regression estimator.

1 Start with the estimand

Let $y_j \in \{0,1\}$ be the reference label of item j, for $j = 1, \ldots, k$. For fixed weights w_j, the target is

$$\theta = w^\mathsf{T} y = \sum_{j=1}^{k} w_j y_j. \tag{1}$$

The weights can be positive for a total or signed for a comparison. They may come from a matrix that connects reusable labels to people, places, or records. The results below apply to quantities linear in the labels, conditional on that matrix and any other inputs used to construct w. Section 2 discusses nonlinear downstream statistics built from the same aggregates.

Browsing supplies one such matrix. Let $C = (c_{ij})$ contain visits by n users to k domains, with each domain serving as an item. The true exposure vector is $T = Cy$. For declared user weights a, the corresponding weights are

$$\theta = a^\mathsf{T} T = a^\mathsf{T} C y = w^\mathsf{T} y, \qquad w = C^\mathsf{T} a. \tag{2}$$

For a difference of group means, $a_i = 1/n_A$ in group A and $a_i = -1/n_B$ in group B. A single user's total sets that user's weight to one and all others to zero. For exposure shares, first replace C by $\mathrm{diag}(V_1^{-1}, \ldots, V_n^{-1})C$, where $V_i = \sum_j c_{ij} > 0$; the same mapping then gives the domain weights for the chosen share comparison. Replacing visits by duration changes the exposure unit but not the mapping.

Regression coefficients are also fixed linear functionals of the labels. For a full-rank demographic design X, write $A = (X^\mathsf{T} X)^{-1} X^\mathsf{T}$ and define the complete-label coefficient $\beta^* = ACy$. A coefficient contrast $r^\mathsf{T}\beta^*$ has user weights $a = A^\mathsf{T} r$ and domain weights $w = C^\mathsf{T} A^\mathsf{T} r$. These are descriptive coefficients for the specified browsing frame; causal or population interpretations require additional assumptions. Throughout, the item weights and any matrices or regressors used to construct them are fixed. Their uncertainty, if relevant to the research question, requires an additional sampling model. Domain labels also define the browsing construct: they measure visits to domains with a given label, not necessarily consumption of that content on mixed-content sites.

Let $q \in \{0,1\}^k$ contain predicted labels, $q_j = \widehat{y}_j$, and let $e = q - y$. The prediction-based estimate is $\widehat{\theta} = w^\mathsf{T} q$, so its error is exactly

$$\Delta = \widehat{\theta} - \theta = w^\mathsf{T} e. \tag{3}$$

Thus the estimand determines which classification mistakes matter. In the browsing example, a popular domain can dominate an individual's total but cancel from a group comparison if the groups

use it equally. A less popular domain visited disproportionately by one group can be influential for that comparison. An error rate that weights all items equally need not describe either target.

For fixed labels and a fixed fitted classifier, Δ is fixed. If the question instead concerns repeated training and prediction, let $b = \mathrm{E}[q - y]$ and $\Sigma = \mathrm{Cov}(q - y)$ under that process while holding y, w fixed. Then $\mathrm{E}[\Delta] = w^\mathsf{T} b$ and $\mathrm{Var}(\Delta) = w^\mathsf{T} \Sigma w$. Shared items and correlated classifier errors enter through these weights and covariances. They require a model of repeated prediction; the count bounds and probability audit below condition on the realized predictions instead. Appendix D gives the covariance details and a rare-error example showing why many items alone do not justify a normal approximation.

2 When an aggregate becomes a regression covariate

Counts and fixed-denominator proportions assembled from predicted labels are also common regression covariates. With fixed aggregation matrix C, their complete-label values are $T = Cy$ and their predicted values are $S = Cq$. Each component of T is linear in item labels. A downstream statistic $F(T, Z)$ need not be: its exact plug-in error is $F(Cq, Z) - F(Cy, Z)$, where Z denotes other fixed inputs. A denominator that depends on predicted labels makes even the underlying proportion nonlinear. The sharp count bounds in Section 3 therefore apply directly only when the declared target is linear in y.

For example, let Y be an observed outcome and V a full-rank matrix of fully observed controls, including an intercept. The coefficient that ordinary least squares would report with complete item labels is

$$\beta(y) = \frac{T^\mathsf{T} M_V Y}{T^\mathsf{T} M_V T} = \frac{y^\mathsf{T} C^\mathsf{T} M_V Y}{y^\mathsf{T} C^\mathsf{T} M_V C y}, \qquad M_V = I - V(V^\mathsf{T} V)^{-1} V^\mathsf{T}, \tag{4}$$

when the denominator is positive. Both numerator and denominator change when labels change, so ordinary attenuation is not a general conclusion. The exact error identity and item-audit moment formulas are in Appendix C. Those formulas do not provide an interval for $\beta(y)$: a ratio of estimated moments can be biased, unstable, or undefined.

This setting intersects work on machine-generated regressors (Fong and Tyler, 2021; Battaglia et al., 2025; Kluger et al., 2025). Battaglia et al. (2025) already study classification and aggregation into regression indices. The distinction examined here is a validation-design one: a verified item can inform many units' aggregates, but usually does not reveal any one unit's entire covariate. Methods requiring a sample of complete unit-level covariates need an audit that obtains them or an adaptation to item-level labels. For the coefficient in equation (4), that adaptation must account for item pairs in the quadratic denominator.

A model-based target requires different assumptions. If information $\mathcal{H}$ contains C, V and the prediction inputs, and $p_j = \mathrm{E}[y_j \mid \mathcal{H}]$, then $Cp = \mathrm{E}[T \mid \mathcal{H}]$. Under $Y = V\gamma + \beta T + \varepsilon$ with $\mathrm{E}[\varepsilon \mid \mathcal{H}] = 0$ and a full-rank design, regressing Y on (V, Cp) identifies the population β. Marginal score calibration across items does not establish these conditional means or the exclusion assumption. This population coefficient is also distinct from the fixed-sample complete-label coefficient in equation (4).

3 What a confusion matrix identifies

A confusion matrix can be exactly correct for the target items and still fail to determine a downstream estimate. This is an information problem before it is a sampling problem. Fix the predictions q and the item weights w in equation (1). We now ask what a validation summary establishes about the

fixed target $\theta = w^\mathsf{T} y$, without specifying a stochastic model for how the classifier's mistakes were generated.

Let $P = \{j : q_j = 1\}$ and $N = \{j : q_j = 0\}$. Suppose the entire target-population confusion matrix is known: f predictions in P are false positives and g predictions in N are false negatives. This grants more information than ordinary held-out validation, whose error rates require a transport assumption. For a finite collection B of weights, let $S_-(B, m)$ and $S_+(B, m)$ denote the sums of its m smallest and largest elements, with a zero sum when $m = 0$.

Proposition 1 (Sharp bounds from confusion counts)**.** *Across binary label vectors compatible with q, f, g, the smallest and largest possible errors $\Delta = \widehat{\theta} - \theta$ are*

$$\begin{aligned} L &= S_-(w_P, f) - S_+(w_N, g), \\ U &= S_+(w_P, f) - S_-(w_N, g). \end{aligned} \tag{5}$$

Consequently $[\widehat{\theta} - U, \widehat{\theta} - L]$ is the smallest interval containing every compatible value of θ. Both endpoints are attainable.

Proof. Only false positives and false negatives contribute: $\Delta = \sum_{j \in \mathrm{FP}} w_j - \sum_{j \in \mathrm{FN}} w_j$. Select f indices from P and g from N. The two selections have disjoint supports and no other restrictions. Selecting the smallest positive-stratum weights and largest negative-stratum weights minimizes the error; reversing these selections maximizes it. Assign the selected indices the corresponding true labels to attain each endpoint. □

The compatible values form a finite set, so the interval is its convex hull; not every interior value need be attainable. A reported sign is established by these counts only if the bounding interval excludes zero. The calculation requires sorting, rather than a general integer-programming solver.

For example, take $q = (1, 1, 0, 0)$, $w = (1, -1, 1, -1)$, and $f = g = 1$. The same confusion matrix permits errors of -2, 0, and 2. Prevalence, accuracy, FPR, and FNR are identical across those assignments. Adding correctly classified items with zero weight makes accuracy arbitrarily close to one without resolving the target. In the browsing application, these can be domains that affect individual exposure but cancel from a group contrast.

Proposition 2 (When counts suffice)**.** *In a stratum of M items sharing the same predicted label, with exactly m errors, the error sum is point identified if and only if $m \in \{0, M\}$ or all its contrast weights are equal. Its identification width obeys*

$$S_+(w, m) - S_-(w, m) \le \min(m, M - m)\{\max_j w_j - \min_j w_j\}. \tag{6}$$

Widths add across strata. For several comparisons, counts determine every comparison if and only if their vector of item weights is constant within each stratum with an interior error count.

Proof. Constant weights, no errors, and all errors each determine the sum. If $0 < m < M$ and two weights differ, choose an error set containing one of those indices but not the other. Exchanging them changes the sum, proving necessity. The difference between two sums of m terms is at most m times the range. Taking complements gives the corresponding bound with $M - m$ terms. The vector statement applies the exchange argument to each coordinate. □

For a stratum mixing predicted labels, the corresponding criterion applies to the signed contributions $(2q_j - 1)w_j$, not to w_j alone. The result specifies what a finer validation table must preserve. Stratifying by predicted label alone is sufficient only when the contrast attaches equal

weights within each partially erroneous stratum. Partitioning items by similar contrast weights can reduce ambiguity; bins chosen for one comparison need not work for another. In browsing, error rates conditional on user demographics are not automatically available: the same domain can enter many groups.

4 What a target-item audit can certify

A probability audit can recover information that a confusion matrix discards. The construction uses established finite-population counting and audit ideas (Stark, 2008; Shekhar et al., 2023). Partition the target items using information known before labels are audited, then draw a fixed-size random sample without replacement within each stratum. Exact hypergeometric inversion gives a confidence interval $[\ell_h, u_h]$ for the number of incorrect predictions in stratum h; allocate error probabilities across strata so their sum is at most δ.

Let O contain the audited items and $\mathcal{Y}_A$ the binary label vectors that match their verified labels and whose error counts in each stratum lie in $[\ell_h, u_h]$. Its construction and the sorting formula for its extremes are in Appendix E.

Proposition 3 (A simultaneous count certificate)**.** *For any fixed item weights w, the interval $[\min_{z\in\mathcal{Y}_A} w^\mathsf{T} z, \max_{z\in\mathcal{Y}_A} w^\mathsf{T} z]$ covers the fixed target $w^\mathsf{T} y$ with probability at least $1-\delta$ over the audit design. The same event covers every linear contrast of this one label vector. Its endpoints are sharp given the audited labels and the count intervals.*

The certificate is simultaneous because the count event constrains the same label vector for every w. It can nevertheless be wide: it retains error counts but discards the weights attached to unaudited error locations. The example and full audit comparison in Appendix E show this loss. Those simulations evaluate a design chosen for one coefficient; they do not establish efficient simultaneous auditing for other contrasts.

For one declared contrast, a stratified difference estimator instead keeps the audited item identities and their weighted residuals:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j\in O_h} w_j(y_j - q_j). \tag{7}$$

With fixed predictions and verified labels, this estimator is design unbiased. A normal interval needs a separate approximation. The conservative finite-sample Hoeffding interval in the appendix is wide in this application; the paper does not establish an efficient finite-sample interval or an optimal audit allocation. The distinction here is between preserving the locations of audited errors and reducing them to counts.

5 What calibrated scores would add

The bounds from exact target-item confusion counts and the probability audit above condition on fixed predictions and labels. Neither requires calibrated classifier scores. The audit needs a known probability-sampling design for target items and verified labels for the sampled items. If the error counts come instead from a separate validation collection, applying them to the target items requires a transport assumption, whether or not that collection's scores are calibrated.

A score-based plug-in uses a different source of information. Let $p_j \in [0,1]$ be a score intended to represent the probability that target item j has label one, and put $\theta_p = w^\mathsf{T} p$. To interpret this as a

conditional mean, introduce a model in which the labels are random given information $\mathcal{I}$ containing the scores and the target weights. Write $\mu_j = \mathrm{E}[y_j \mid \mathcal{I}]$. Then

$$\mathrm{E}[\theta_p - \theta \mid \mathcal{I}] = w^{\mathsf{T}}(p - \mu). \tag{8}$$

The plug-in equals $\mathrm{E}[\theta \mid \mathcal{I}]$ if and only if this weighted error is zero. A sufficient condition is $p_j = \Pr(y_j = 1 \mid \mathcal{I})$ for each target item. For a particular target, offsetting score errors can also make the weighted sum zero without item-level conditional calibration.

Ordinary calibration, $\Pr(y = 1 \mid p = s) = s$ across items, is weaker. For example, let half the items have label one and weight one, and the other half have label zero and weight zero. Give every item score $1/2$. The scores are empirically calibrated in the unweighted item collection, but θ_p is only half of the realized target. Conditioning on features that distinguish the two weight groups, or directly checking the weighted residual, exposes the failure. Calibration assessed on a separate collection also has to transfer to the target items and their label definition.

For a browsing visit share, $w_j = v_j / \sum_\ell v_\ell$ gives $\theta_p = \sum_j v_j p_j / \sum_j v_j$. Conditional probabilities for the specified content label would make this the model's expected visit share. A group contrast or regression coefficient has different, possibly signed weights; its required weighted score condition changes accordingly.

Even valid target-item conditional probabilities identify a conditional expectation, not the realized finite-item target. Under the conditional-label model, uncertainty around that expectation depends on the joint distribution:

$$\mathrm{Var}(\theta \mid \mathcal{I}) = w^{\mathsf{T}} \mathrm{Cov}(y \mid \mathcal{I}) w. \tag{9}$$

Item-level calibration supplies neither the off-diagonal covariances nor, by itself, a coverage guarantee for an interval about the realized target. Estimating the scores introduces further uncertainty. These model-based claims are distinct from the fixed-label, design-based coverage of the audit.

6 A browsing-data stress test

The empirical question is how much information confusion counts retain for comparisons supported by an actual browsing matrix. The fixed benchmark contains 1,132 pseudonymized users, 34,078 domains, 103,853 user–domain records, and 4,767,099 visits. It is the bundled `fewlab` extract of the June 2022 YouGov Pulse/RealityMine panel with Blacklight scanner outcomes. The replication package pins the source commit and file hashes. Its analytic universe contains domains with recorded scanner labels; exposure shares use visits within that universe as their denominator. These recorded outcomes are reference labels for the experiment, not error-free measures of all tracking activity. The exercise makes no causal or population-representative claim about demographic differences.

We retain the extract's demographic design matrix: an intercept and indicators for gender, race, education, and source-coded age groups. Omitted categories are Male, White, HS or Below, and under 25. For each of seven scanner outcomes, we calculate all twelve non-intercept least-squares coefficients for both visit totals and visit shares. Each coefficient is a fixed weighted label sum. Two scanner columns are counts and are dichotomized at a positive value; the other five are binary. Complete labels, valid joins, positive visits, and full column rank are checked before estimation.

Predictions are imposed, rather than produced by a fitted classifier. Within each true class, we randomly flip the integer part of 1%, 5%, or 10% of the labels. We then perform two distinct exercises. First, holding those predictions fixed, Proposition 1 gives the true-coefficient range compatible with their exact confusion counts. Second, holding the reference labels fixed, we rearrange the same

numbers of false positives and false negatives to minimize or maximize the predicted coefficient. The latter asks whether two classifiers with identical confusion matrices can report opposing comparisons. It is a worst-case sensitivity analysis, not an estimate of likely classifier bias. Extrema are computed separately for each coefficient; a single error arrangement need not reverse all comparisons at once.

Table 1: What exact confusion counts establish at 1% imposed error in each true class

Scanner outcome	Sign identified	Median realized error (pp)	Median bound width (pp)
Advertising trackers	0/12	0.43	47.8
Third-party cookies	0/12	0.09	52.5
Canvas fingerprinting	0/12	0.34	29.5
Session recording	0/12	0.40	31.6
Key logging	0/12	2.10	21.3
Facebook pixel	0/12	0.10	43.6
Google Analytics	0/12	0.33	21.6

Each row covers twelve demographic coefficients of visit shares in the fixed browsing frame. "Sign identified" counts bounding intervals excluding zero for fixed random predictions. Realized error is the median absolute difference between the prediction-based and reference coefficients. Bound width is the median identification-interval width; both are in percentage points (pp). Error counts are rounded down within each true class. These are not sampling intervals. All outcomes, coefficients, scales, and error fractions are in the machine-readable results.

At the smallest imposed error fraction, none of the 84 share-coefficient signs is identified by the full confusion matrix (Table 1). The realized errors from random placement are usually much smaller than the count-compatible ranges. For example, the reference female coefficient for Facebook-pixel visit share is 0.828 percentage points. The fixed random predictions differ by 0.040 percentage points, while their count-compatible interval for the reference coefficient is $[-17.00, 16.72]$ percentage points. Moving the same numbers of errors can reverse all 84 share-coefficient signs, but those placements are adversarial. These results show what the confusion counts fail to rule out for this frame, not what a fitted classifier is likely to do. Total coefficients and larger imposed error fractions are in the replication results.

An audit can retain more. Appendix E evaluates count projection and direct weighted residual estimation on the same probability-audit draws for one declared coefficient. The count certificate covered that coefficient in every draw but never settled its sign. The direct normal intervals were narrower but undercovered in the smallest random design; the finite-sample Hoeffding baseline was wider than count projection. The exercise illustrates the information lost by counts and the limits of these particular intervals. It does not compare efficient finite-sample procedures or assess the theorem's simultaneous utility across the other coefficients.

7 Observed disagreement between domain classifiers

The imposed-error experiment controls the number and locations of mistakes. A separate comparison uses observed outputs from the adult-domain study of Sood and Shen (2024). Its public domain summary records 6,236,834 visits to 64,074 domains in the June 2022 YouGov panel. The study used a Piedomains rule that marks a domain when its predicted class is "porn" and its reported class probability is at least 0.033. We compare that rule with recorded "porn" categories from Bitdefender, "sex" from Forcepoint, and "Pornography" from alphaMountain. Each row retains only domains with a recorded category from that provider, and visit shares use the visits in that row's frame as their denominator. Missing provider categories are not negative labels.

Table 2: Observed classifier disagreement and visit-weighted shares

Reference	Domains	Visits (%)	Agree (%)	Domain + (%)	Visit + (%)	Count range (%)
Bitdefender	43,391	84.2	97.7	1.92/3.36	1.05/1.71	[0.07, 82.11]
Forcepoint	47,950	87.5	98.0	1.65/3.04	1.02/1.64	[0.07, 80.61]
alphaMountain	36,063	52.7	99.0	0.98/1.35	1.38/2.16	[0.04, 68.07]

Piedomains is the proxy in every row; the named provider supplies the reference category. "Visits" is the share of all 6,236,834 visits represented by domains with a recorded reference category. Agreement counts domains equally. "Domain +" and "Visit +" list Piedomains/provider percentages, first for domains counted equally and then for visits within that row's frame. The count range is the sharp range of the provider visit share if only the two directional disagreement counts are kept, holding Piedomains labels and visit weights fixed. The providers use different category definitions; none is an adjudicated truth label. A recorded category other than the named positive one counts as negative for this comparison, including nonspecific categories such as "misc."

On the 43,391 domains categorized by Bitdefender, which account for 84.2% of visits, the two classifications agree on 97.7% of domains. Piedomains marks 1.92% of domains positive, compared with 3.36% under Bitdefender's category. Their visit-weighted shares are 1.05% and 1.71%, respectively. The discrepancy is visible before weighting domains by visits, so this comparison does not isolate an effect of visit concentration. There are 193 Piedomains-only and 816 Bitdefender-only positive domains.

With both classifications observed, the provider's visit share is known. If the disagreement locations were discarded and only their directional counts retained, its compatible range would be [0.07, 82.11]%. A single Piedomains-negative domain carries 18.8% of visits in this frame and can dominate the count-only upper bound. Appendix F shows how that range changes when the observed label pairs of the most-visited domains are retained. This isolates information discarded by a confusion table; it does not establish that the extreme allocations are plausible classifier errors.

This is a comparison between imperfect classification systems, not an estimate of either system's error against verified content labels. The public repository files used here do not include the person–domain matrix needed to apply the same calculation to the adult study's partisan coefficients. A probability audit of domains in that matrix, with an explicit content-labeling rule, would support a direct validation analysis of those coefficients without assuming calibrated scores. The reported Piedomains class probability and the provider categories alone do not establish the conditional probabilities required for the score plug-in in Section 5.

8 Implications for validation

For a declared linear target, exact confusion counts constrain the number of mistakes but generally not their weighted sum. Their sharp bounds state what those counts alone establish. A target-item

probability audit can add design-based coverage; retaining sampled weighted residuals allows direct estimation for a chosen target. If several targets matter, their different weight vectors determine which error locations a common audit must preserve.

These fixed-label procedures need verified labels and, for an audit, a known sampling design. They do not require calibrated scores. Applying error rates from another collection requires transport assumptions; interpreting scores as conditional probabilities requires stronger conditions described in Section 5. For aggregated predicted covariates, estimating their components does not by itself give valid inference for a regression slope. The browsing-specific common-rate correction and its assumptions appear in Appendix A.

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Appendix

A Bias and the denominators of error rates

Let α_j be the probability of predicting one for a truly negative domain and β_j the probability of predicting zero for a truly positive domain, conditional on the browsing design. Then

$$b_j = \alpha_j(1 - y_j) - \beta_j y_j, \qquad \mathrm{E}[d_i \mid C, y] = \sum_j c_{ij}\{\alpha_j(1 - y_j) - \beta_j y_j\}. \tag{A1}$$

False-positive rates condition on true negatives; false-negative rates condition on true positives. Multiplying a false-positive rate by predicted positives uses the wrong denominator. Such a calculation would instead require the probability of a true negative conditional on a positive prediction, together with assumptions that justify transferring that probability to the weighted population. These conditional probabilities are not interchangeable.

If $\alpha_j = \alpha$ on negative domains and $\beta_j = \beta$ on positive domains,

$$\mathrm{E}[d_i \mid C, y] = \alpha(V_i - T_i) - \beta T_i = V_i\{\alpha - (\alpha + \beta)\pi_i\}, \qquad \pi_i = T_i/V_i, \tag{A2}$$

where the share π_i is defined for $V_i > 0$. At a fixed true share, expected bias scales with browsing volume unless the expression in braces is zero. For $\alpha + \beta > 0$, cancellation occurs at $\pi_i = \alpha/(\alpha + \beta)$. Equal positive error rates cancel only when half of visits are to truly positive domains. Thus the difference between FPR and FNR alone does not determine bias. Nor does the number of distinct visited domains: repeated visits matter through c_{ij}.

The common-rate assumption is stronger than knowing an average error rate on a validation set of domains. Errors may depend on popularity, domain type, or features associated with a particular group's browsing. In that case, equation (A1), with rates conditional on those features, is the relevant expression. The bias in a share is $\mathrm{E}[d_i \mid C, y]/V_i$; dividing by volume removes the mechanical scale factor, not misclassification bias.

With known α, β and $\alpha + \beta \neq 1$, this model gives an unbiased correction:

$$\widetilde{T}_i = \frac{\widehat{T}_i - \alpha V_i}{1 - \alpha - \beta}, \qquad \mathrm{E}[\widetilde{T}_i \mid C, y] = T_i. \tag{A3}$$

The correction is unstable when the denominator is close to zero and may return values outside $[0, V_i]$. Clipping changes its expectation. Estimating error rates introduces additional uncertainty shared across users; rates estimated on unweighted domains need not transfer to weighted exposure. This is the connection to classifier-based quantification (Vaz et al., 2019), with the additional requirement that the rates apply to each user's weights.

B A connection to label sampling

The same mapping supplies a remedy when a researcher can verify a probability sample of items. The companion `fewlab` project develops sampling designs for reusable labels (Sood, 2026). In its browsing application, the regression decomposition is $H = AC$, with column h_j giving the influence of domain j on the coefficient. For exposure shares, replace C by its row-normalized version. For any scalar target in equation (3), the corresponding item influence is w_j.

Freeze predictions q_j before drawing a label sample. Let R_j indicate selection, with known inclusion probability $s_j > 0$ for every item. Conditional on the fixed population and predictions,

the difference estimator is

$$\widetilde{\theta} = \sum_j w_j q_j + \sum_j \frac{R_j}{s_j} w_j (y_j - q_j). \tag{B1}$$

Since $\mathrm{E}_R[R_j/s_j] = 1$, its expectation over label sampling is $\sum_j w_j y_j = \theta$, regardless of the classifier's errors. Verified labels must measure y_j correctly. Retraining the proxy on the same sampled labels requires a separate argument; the fixed-proxy proof does not cover it.

Under independent Bernoulli item sampling,

$$\mathrm{Var}_R(\widetilde{\theta}) = \sum_j \frac{1 - s_j}{s_j} w_j^2 (y_j - q_j)^2. \tag{B2}$$

The realized sample gives an unbiased variance estimator by replacing each summand with $R_j(1 - s_j)w_j^2(y_j - q_j)^2/s_j^2$. Fixed-size or balanced designs require their joint-inclusion terms or a justified approximation. An inaccurate proxy can increase variance relative to omitting it, but does not introduce design bias under these assumptions. This is uncertainty from randomized label acquisition, with y and q fixed; it is distinct from the prediction-error process in equation (D4). Both calculations preserve the reuse of an item label across downstream records.

C Auditing an aggregated regression covariate

The general plug-in error is $F(Cq, Z) - F(Cy, Z)$. If scalar F is differentiable throughout the segment from Cy to Cq, it also has the exact representation

$$F(Cq, Z) - F(Cy, Z) = \int_0^1 \left[C^\mathsf{T} \nabla_T F\{Cy + sC(q - y), Z\} \right]^\mathsf{T} (q - y)\, ds. \tag{C1}$$

Unlike the fixed weights of a linear target, these effective item weights depend on unknown labels along the path. Thresholds need not be differentiable.

For the OLS coefficient in equation (4), set $d = C(q - y)$, $t = M_V C y$, $h = M_V d$, and $u = M_V Y - \beta(y)t$, so $t^\mathsf{T} u = 0$. If the predicted-regressor denominator is positive, the exact error is

$$\beta(q) - \beta(y) = \frac{h^\mathsf{T} u - \beta(y)\{t^\mathsf{T} h + h^\mathsf{T} h\}}{(t + h)^\mathsf{T}(t + h)}. \tag{C2}$$

The quadratic term can remain when first-order terms cancel, and the coefficient can move in either direction.

The fixed-sample coefficient in equation (4) has a linear numerator and a quadratic denominator in the item labels. Put $a = C^\mathsf{T} M_V Y$, $G = C^\mathsf{T} M_V C$, and $r = y - q$. Then

$$\begin{aligned} N &= y^\mathsf{T} a = q^\mathsf{T} a + r^\mathsf{T} a, \\ D &= y^\mathsf{T} G y = q^\mathsf{T} G q + 2 r^\mathsf{T} G q + r^\mathsf{T} G r. \end{aligned} \tag{C3}$$

Let R_j indicate a domain's inclusion in a probability audit, with $\pi_j = \Pr_R(R_j = 1) > 0$ and $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1) > 0$ for every relevant pair. Set $\pi_{jj} = \pi_j$. The following estimates are design unbiased for the two moments, conditional on C, V, Y, y, q:

$$\begin{aligned} \widehat{N} &= q^\mathsf{T} a + \sum_j \frac{R_j}{\pi_j} r_j a_j, \\ \widehat{D} &= q^\mathsf{T} G q + 2 \sum_j \frac{R_j}{\pi_j} r_j (Gq)_j + \sum_{j,\ell} \frac{R_j R_\ell}{\pi_{j\ell}} r_j G_{j\ell} r_\ell. \end{aligned} \tag{C4}$$

The identities follow from $\mathrm{E}_R[R_j/\pi_j] = 1$ and $\mathrm{E}_R[R_j R_\ell/\pi_{j\ell}] = 1$. No independent-item-error or calibrated-score assumption is used. The ratio $\widehat{N}/\widehat{D}$ is generally not design unbiased and can be unstable or undefined when the estimated denominator is near zero or nonpositive. Valid inference for the ratio requires additional design analysis; its variance involves inclusion probabilities for sets of up to four items. The formulas identify the information a domain audit would have to preserve. The linear-label difference estimator in equation (B1) does not by itself correct a regression that uses Cy as a covariate.

D Repeated-prediction error and dependence

For a fixed fitted classifier and fixed labels, Δ is fixed. To discuss bias and variance, specify a repeated-prediction process, such as repeated training samples, while fixing y, w and the inputs that determine w. All moments below condition on this fixed design. Define $b = \mathrm{E}[e]$ and the full joint covariance $\Sigma = \mathrm{Cov}(e)$, without assuming independence across items. Then

$$\begin{aligned}
\mathrm{Bias}(\widehat{\theta}) &= w^{\mathsf{T}} b, \\
\mathrm{Var}(\Delta) &= w^{\mathsf{T}} \Sigma w = \sum_j w_j^2 \Sigma_{jj} + 2 \sum_{j<\ell} w_j w_\ell \Sigma_{j\ell}, \qquad \text{(D1)} \\
\mathrm{E}[\Delta^2] &= (w^{\mathsf{T}} b)^2 + w^{\mathsf{T}} \Sigma w. \qquad \text{(D2)}
\end{aligned}$$

Equations (1)–(D2) connect the research quantity to its item weights, systematic error, and uncertainty. Covariance between item errors enters the estimate's variance. Dependence alone does not determine bias; systematic errors on items with nonzero weights can make $w^{\mathsf{T}} b$ nonzero.

Browsing totals and regression vectors are corresponding joint quantities. Writing $\widehat{T} = Cq$, $d = \widehat{T} - T$, and $\widehat{\beta} = ACq$ gives

$$\begin{aligned}
d &= Ce, & & \qquad \text{(D3)} \\
\mathrm{E}[d] &= Cb, & \mathrm{Cov}(d) &= C\Sigma C^{\mathsf{T}}, \qquad \text{(D4)} \\
\mathrm{E}[\widehat{\beta} - \beta^*] &= ACb, & \mathrm{Cov}(\widehat{\beta} - \beta^*) &= AC\Sigma C^{\mathsf{T}} A^{\mathsf{T}}. \qquad \text{(D5)}
\end{aligned}$$

The same item error is reused across all visits and users. These formulas quantify the specified prediction-error process; they do not include browsing-log error, user sampling, or uncertainty about the construct. They concern predicted exposure as an outcome. Noisy exposure as a regressor is a different errors-in-variables problem treated in Section 2.

D.1 Dependence follows the item weights

There are two sources of dependence. An item's predicted label can be reused across records, and different items can be classified by the same fitted model. Shared training data, model parameters, and similar item features can cause errors to move together under repeated fitting. Distinct items do not establish independent errors. In browsing, users need not visit the same domains to have correlated measurement errors: errors on different visited domains can covary. In equation (D1), positively correlated mistakes increase variance when their item weights have the same sign, and decrease it when the weights have opposite signs.

For a limiting example, suppose centered errors are shared within item families: $e_j - b_j = u_{g(j)}$, with independent family shocks u_g of variance τ_g^2. This is compatible with binary predictions when the classifier assigns one common random prediction to every item in a family. Then

$$\mathrm{Var}(\Delta) = \sum_g \tau_g^2 \left(\sum_{j:g(j)=g} w_j \right)^2. \qquad \text{(D6)}$$

A family of m items with unit weights contributes $m^2 \tau_g^2$, whereas assuming independent item errors would give $m\tau_g^2$. Signed weights can instead cancel within a family. The relevant concentration is in aggregate family influence, not just individual item influence. This example is a special covariance structure, not an assertion that the right families are known or that all within-family mistakes are perfectly correlated.

Independent item errors are another special case: Σ is diagonal, with entries v_j. For browsing totals, equation (D4) reduces to

$$\mathrm{Var}(d_i) = \sum_j c_{ij}^2 v_j, \qquad \mathrm{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} v_j. \tag{D7}$$

Let α_j be the false-positive probability on a truly negative item and β_j the false-negative probability on a truly positive item. For binary predictions, $v_j = \alpha_j(1-\alpha_j)$ on negative items and $v_j = \beta_j(1-\beta_j)$ on positive items. In browsing, repeated visits multiply one classification error; treating visits as independent would incorrectly replace squared visit weights by visit counts. Even this special case induces dependence across users who share uncertain domains.

An uncertainty procedure must follow the same structure as the estimand. Resampling downstream records while holding predictions fixed does not propagate repeated-training uncertainty. Each replicate must preserve reused labels across records and the joint distribution of errors across items. Independently perturbing each item label imposes a diagonal Σ; resampling the classifier fit and predicting all items together can preserve dependence induced by that fitting process, provided the resampling scheme represents that process. A prediction-error covariance cannot simply be added to an arbitrary reported regression covariance: the components may overlap or be dependent. Neither larger downstream samples nor clustering records alone guarantees that this error disappears, and standard errors do not repair bias.

D.2 Many items do not ensure a normal approximation

Even with independent item errors, diffuse variance shares alone do not justify Gaussian inference. Let all true labels be zero and let each of k items be falsely labeled positive independently with probability $1/k$. With $w_j = 1$, total error B_k is binomial with mean one and variance $1 - 1/k$. The largest item variance share is $1/k$, yet $B_k - 1 \Rightarrow \mathrm{Poisson}(1) - 1$, not a normal distribution. At $k = 10{,}000$, an oracle interval using the exact bias and variance but a normal 1.96 cutoff has coverage 91.97%. This calculation uses exact binomial probabilities, not simulation.

For independent bounded binary errors, a sufficient Lindeberg condition is

$$\frac{\max_j |w_j|}{\sqrt{\sum_j w_j^2 v_j}} \to 0.$$

A small maximum variance share $\max_j w_j^2 v_j / \sum_j w_j^2 v_j$ is weaker when errors become rare. The example violates the former condition and satisfies the latter. A growing number of items cannot substitute for a condition on the distribution of their errors. This is a prediction-error example, distinct from the fixed-label audit design.

E Count-certificate construction and audit diagnostics

Partition items into strata $h = 1, \dots, H$ before observing their audited labels. Sample n_h of the M_h items without replacement. If K_h is the unknown number of incorrect predictions and X_h the sampled error count, then $X_h \sim \text{Hypergeometric}(M_h, K_h, n_h)$. Assign nonnegative error budgets δ_h with $\sum_h \delta_h \le \delta$. Retain integers K for which both $\Pr_K(X_h \le x_h) \ge \delta_h/2$ and $\Pr_K(X_h \ge x_h) \ge \delta_h/2$. Their extrema are ℓ_h, u_h. A census stratum has $\ell_h = u_h = x_h$; an unaudited stratum contributes no count restriction.

Let O be the audited items, U_h the unaudited items in stratum h, $c_j = (2q_j - 1)w_j$, and $d_O = \sum_{j \in O} w_j(q_j - y_j)$. The smallest and largest compatible errors are

$$L_A = d_O + \sum_h \min_{\ell_h - x_h \le m \le u_h - x_h} S_-(c_{U_h}, m),$$
$$U_A = d_O + \sum_h \max_{\ell_h - x_h \le m \le u_h - x_h} S_+(c_{U_h}, m). \tag{E1}$$

The optimization is over integers. With signed weights, an extreme need not occur at the largest admissible error count.

Proof of Proposition 3. Inverting each pair of hypergeometric tails excludes the true K_h with probability at most δ_h. A union bound therefore includes all true counts with probability at least $1-\delta$. On that event, the true errors in U_h number between $\ell_h - x_h$ and $u_h - x_h$. Sorting and choosing a count in that range gives the extrema in equation (E1); the choices are independent restrictions across strata and hence jointly attainable. The true label vector lies in the same configuration set for every w, so the coverage event applies simultaneously to all its linear contrasts. □

The labels and predictions are fixed under this design. Systematic or correlated classifier mistakes do not invalidate coverage, which is over repeated audit samples. The count region can be conservative, and sharp projection of that region does not make its interval shortest. The guarantee concerns one label vector; combining separate audits of several outcomes needs an additional error allocation. Optional stopping, adaptive strata, and imperfect audit labels require other procedures or assumptions.

Information lost by count projection. Let all N predictions equal one, half the weights equal $1/N$ and half equal $-1/N$. Suppose a fraction $p < 1/2$ of labels in each half are wrong. The true contrast is zero. If $n \to \infty$ but $n/N \to 0$, an unstratified audit consistently learns the overall error fraction. Yet unaudited errors can still be placed almost entirely on either weight sign: the projected interval width converges to $2p$. A direct estimate of the weighted audit residual instead has standard error of order $n^{-1/2}$. Separating the two weight signs into strata removes this particular information loss.

For a declared contrast, the stratified difference estimator uses $z_j = w_j(y_j - q_j)$ directly:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j \in O_h} z_j, \qquad \widehat{V} = \sum_h \frac{M_h^2(1 - n_h/M_h)}{n_h} s_{z,h}^2. \tag{E2}$$

Under independent stratum draws, the estimate and variance estimator are design unbiased when each noncensus stratum has at least two sampled items. Normal intervals need a separate approximation. A conservative finite-sample alternative applies Hoeffding's without-replacement inequality (Hoeffding, 1963) to each bounded residual mean and sums the stratum radii. More efficient weighted audit procedures exist (Shekhar et al., 2023).

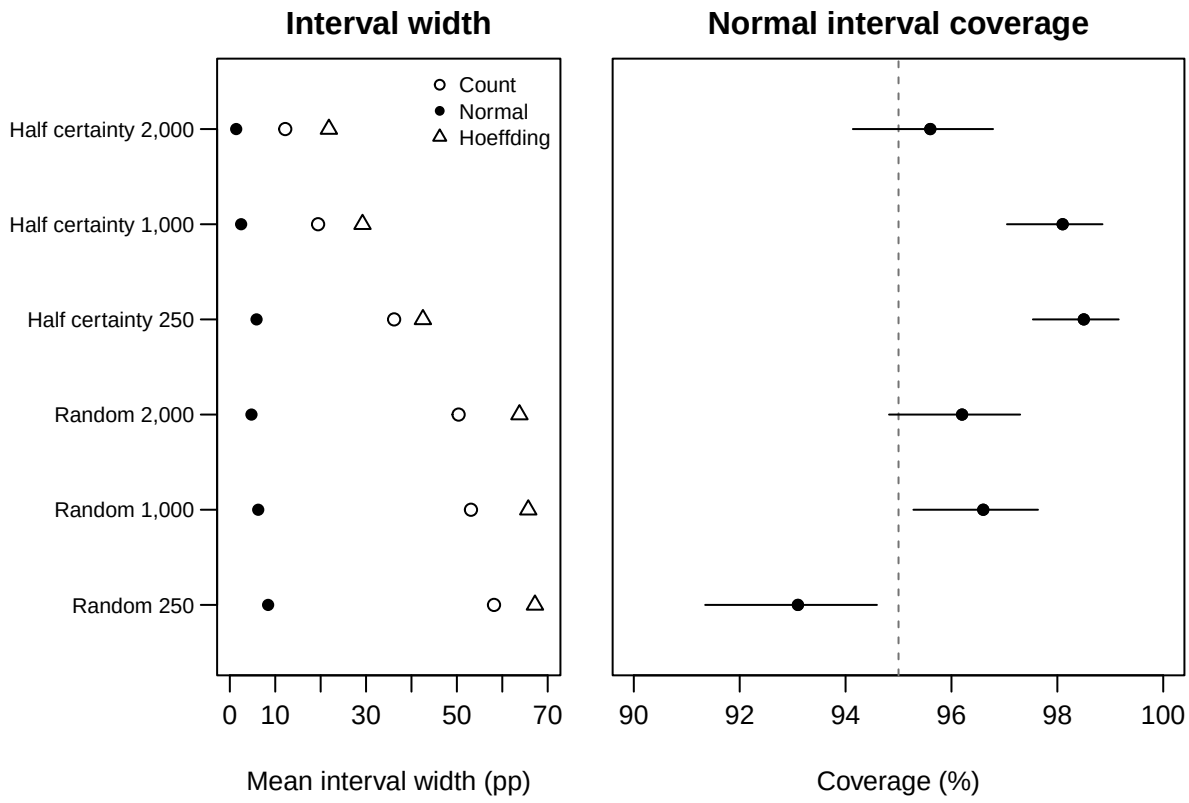


Figure E1: Width and coverage for the Facebook-pixel female share coefficient. Rows compare audit budgets and designs in the fixed browsing frame. Left: mean interval width in percentage points over 1,000 audits. Right: normal interval coverage, with exact 95% binomial Monte Carlo intervals and a dashed 95% reference. Count and Hoeffding intervals covered in every simulated draw. "Half certainty" audits the largest absolute influences with half the budget. Count intervals cover all contrasts of this outcome simultaneously; direct intervals target this coefficient. The randomness is domain auditing; users and predictions are fixed.

Table E1: All audit designs and methods for the focal coefficient

Design	Budget	Interval	Coverage (%)	Width (pp; MCSE)	Correct sign (%)	Wrong sign (%)
Random	250	Count projection	100.0	58.2 (0.06)	0.0	0.0
Random	250	Direct normal	93.1	8.4 (0.50)	39.6	0.0
Random	250	Direct Hoeffding	100.0	67.2 (0.01)	0.0	0.0
Random	1,000	Count projection	100.0	53.1 (0.05)	0.0	0.0
Random	1,000	Direct normal	96.6	6.3 (0.19)	32.9	0.0
Random	1,000	Direct Hoeffding	100.0	65.7 (0.02)	0.0	0.0
Random	2,000	Count projection	100.0	50.4 (0.05)	0.0	0.0
Random	2,000	Direct normal	96.2	4.8 (0.11)	34.4	0.0
Random	2,000	Direct Hoeffding	100.0	63.8 (0.03)	0.0	0.0
Half certainty	250	Count projection	100.0	36.2 (0.05)	0.0	0.0
Half certainty	250	Direct normal	98.5	5.9 (0.29)	40.6	0.0
Half certainty	250	Direct Hoeffding	100.0	42.5 (0.00)	0.0	0.0
Half certainty	1,000	Count projection	100.0	19.4 (0.04)	0.0	0.0
Half certainty	1,000	Direct normal	98.1	2.5 (0.04)	41.9	0.0
Half certainty	1,000	Direct Hoeffding	100.0	29.2 (0.00)	0.0	0.0
Half certainty	2,000	Count projection	100.0	12.2 (0.03)	0.0	0.0
Half certainty	2,000	Direct normal	95.6	1.4 (0.01)	65.0	0.0
Half certainty	2,000	Direct Hoeffding	100.0	21.8 (0.00)	0.0	0.0

Each row summarizes 1,000 domain audits. Width is the mean in percentage points, with its Monte Carlo standard error in parentheses. Sign columns report exclusion of zero in the correct or wrong direction. All methods within a design use the same draws. Exact binomial Monte Carlo intervals and replicate results accompany the package.

The random and half-certainty designs were chosen for one coefficient. They do not establish that this design is efficient for other contrasts. The count certificate covered the target in every simulated draw but never excluded zero. Direct normal intervals were much narrower, yet coverage in the smallest random design fell below the nominal level. The finite-sample Hoeffding baseline was wider than count projection. These comparisons do not identify a finite-sample-valid, efficient direct interval for this design.

Table E2: Refining validation strata for the focal coefficient

Strata	Exact-count width (pp)	Audit width (pp; MCSE)	Coverage (%)
2	49.7	53.2 (0.05)	100.0
10	34.7	47.1 (0.05)	100.0
50	21.8	48.2 (0.03)	100.0

Rank bins of signed influence cross predicted labels. Exact-count width conditions on the population count in every stratum; audit width averages 95% count-projection intervals over 1,000 domain samples of size 1,000. Width uncertainty is its Monte Carlo standard error. These partitions were examined after the initial audit results. No design certified the sign.

The narrower exact-count range from finer strata need not translate into a narrower audit interval: the fixed sample is divided among more strata and the error budget is split across their discrete hypergeometric intervals. The 10- versus 50-stratum mean-width difference should be read with the reported Monte Carlo uncertainty, not as evidence of an optimal partition.

F Retaining the highest-visit domain labels

Table F1 repeats the count-information exercise after retaining both observed classification labels for the highest-visit domains. For the remaining domains it retains only the numbers positive under Piedomains alone and under the provider alone. The ordering uses visit count with domain name to break ties. The count range is sharp under exactly this information: the head contribution is fixed, while the tail discrepancies can occupy any compatible tail domains. At zero retained domains it equals the range in Table 2. The provider's observed visit share lies inside every range.

Table F1: Count-compatible provider visit shares after retaining label pairs for the highest-visit domains

Provider	Domains retained	Visits retained (%)	Count range (%)	Width (pp)
Bitdefender	0	0.0	[0.07, 82.11]	82.05
Bitdefender	10	42.2	[0.07, 40.06]	39.99
Bitdefender	50	56.8	[0.07, 25.80]	25.73
Bitdefender	250	71.5	[1.00, 13.65]	12.64
Bitdefender	1,000	83.4	[1.33, 5.98]	4.64
Forcepoint	0	0.0	[0.07, 80.61]	80.54
Forcepoint	10	40.6	[0.07, 40.16]	40.09
Forcepoint	50	55.1	[0.07, 25.99]	25.92
Forcepoint	250	69.6	[0.86, 14.01]	13.15
Forcepoint	1,000	81.9	[1.24, 6.24]	4.99
alphaMountain	0	0.0	[0.04, 68.07]	68.03
alphaMountain	10	30.1	[0.04, 38.33]	38.30
alphaMountain	50	49.0	[0.56, 21.37]	20.82
alphaMountain	250	67.2	[1.44, 9.12]	7.68
alphaMountain	1,000	81.5	[1.87, 4.03]	2.16

The percentages use each provider row's categorized-domain visit total as its denominator. "Visits retained" is the share of that total on domains whose two observed labels remain known. Width is in percentage points. The providers use different category definitions and neither label is an adjudicated content truth. Retaining a label pair here is an information exercise, not a claim that a manual audit was performed.